NASA SPACE APPS CHALLENGE 2026 · ADAPTING FARMS WITH NASA DATA

Rotation Explorer

A free, map-based web tool that compares crop rotations for any place on Earth using NASA data, explains the answer in plain words, and always shows how sure it is and what it cannot see.

Project summary · 14 November 2026 · Repository: github.com/MTahaNadeem/NASA-Hacakthon

10
languages, including Urdu and Arabic (right-to-left)

12
crops and 12 rotations, including rice + wheat, cotton + wheat and sugarcane

9
quick-pick places, and any point on Earth by map, search or GPS

6
transparent scoring rules, no black-box AI

7
automated test suites plus browser and accessibility checks

17
commits, all pushed to your GitHub repository

The idea in three sentences

Farmers in places like Punjab, Pakistan often grow the same crop on the same land for years, and the cost shows up slowly in soil, water and yield. The information to judge a change exists in NASA satellite data, but it is scattered, technical and gives no sense of certainty. Rotation Explorer puts it in one place: pick a location, say what matters to you, and read a plain-language comparison with the evidence, the uncertainty and the gaps shown.

Honest status at a glance

Built and tested the whole application, the scoring engine, the data pipeline scripts, 10 languages, sharing, printing and offline use.

Partly done real data: NASA POWER weather is verified for all 10 places, but the data files in the repository still hold only Faisalabad and Ames (weather, no soil). One command, `python pipeline/run_all.py`, builds the rest on your computer.

Still to do load MODIS and SMAP satellite data (needs your NASA login on your machine), deploy a public link, record the 30-second video or slides, have a fluent speaker check the translations, and verify the crop-rule sources.

1. How it works

NASA POWER
20 years of daily rain and temperature

SoilGrids
soil pH, texture and carbon with uncertainty (ISRIC, not NASA)

MODIS · SMAP
vegetation cover and soil moisture (via your Earthdata login)

→

Data pipeline
Python scripts build one small file per place

→

Scoring engine
six readable rules, scored for every historical year

→

Web page
four plain steps, runs in any browser

The app is a static website: it needs no server and no account. For the places in the quick-pick list it can read pre-built files (fast, all NASA sources). For any other point on Earth it fetches NASA POWER and SoilGrids live in the browser, so a farmer can click or search anywhere. The scoring runs entirely in the browser, which is why the sliders update instantly.

The scoring in plain terms

Each rotation is scored from 0 to 1 for every one of the last 20 years on six rules. The three sliders decide how much each group of rules counts: **S = Σ (weight × score)**, with the weights adding up to 1. There is no machine learning, because there is no labelled data on how rotations performed, and a black box could not be checked against anything.

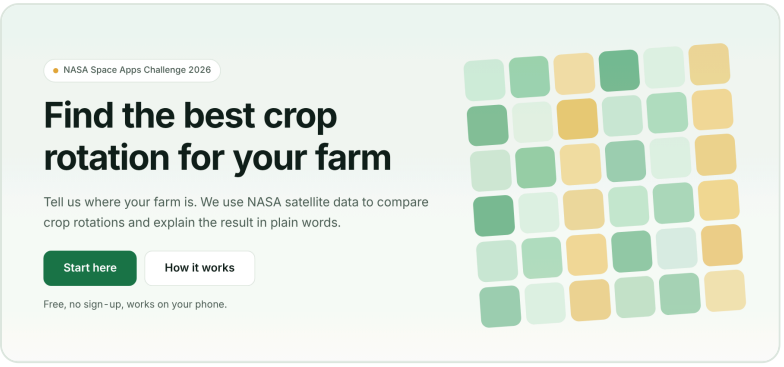
RULE	WHAT IT MEASURES	DATA	SLIDER
Water demand vs. supply	How much of the crop's water need rain covers	POWER rain (+ SMAP moisture)	Water saving
Heat and frost exposure	Heat or frost days in the crop's sensitive months	POWER temperature	Income (yield risk)
Extreme-rain risk	Days with 50 mm or more in sensitive months	POWER (IMERG if loaded)	Income (yield risk)
Bare-field exposure	Months the field is empty and vegetation is low	MODIS NDVI	Soil building
Soil fit	Soil pH and clay against the crop's preferred range	SoilGrids	Soil building
Soil carbon effect	Legumes rebuild carbon; cereals use it up, more so when carbon is low	SoilGrids + crop type	Soil building

- **Ranges, not single numbers.** Each result shows the 10th to 90th percentile across the years, widened by SoilGrids' 5th to 95th percentile soil values. If two rotations' ranges overlap heavily, the app says there is "no clear winner".
- **Confidence** (high, medium, low) drops for close calls, wide year-to-year ranges, missing soil data and fewer than three data sources. With only weather and soil loaded it can never be "high", which is deliberate.
- **If a source is missing, its rule is dropped** and listed under "Data gaps". Nothing is ever invented.
- **Plain-language layer.** A rule scoring 0.7 or more becomes a strength ("Low risk from heavy rain"); 0.5 or less becomes a watch-out ("Needs a lot of extra irrigation water"). The headline "Fit 80/100" is the average score times 100.

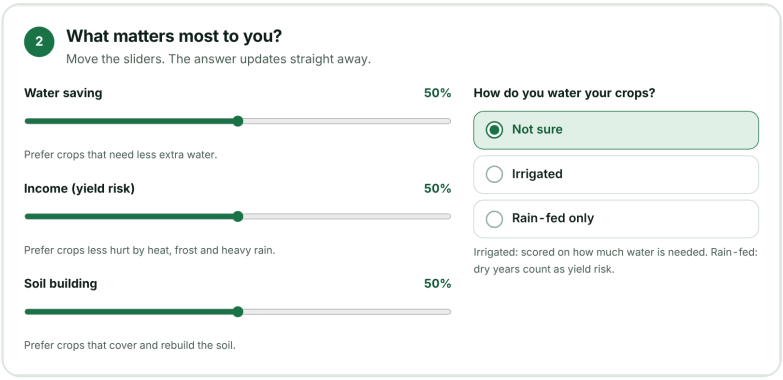
2. What a user sees: four steps

The page was redesigned twice, first as a dark dashboard and then, following your feedback, as a light page written for non-experts. It is now a simple top-to-bottom flow.

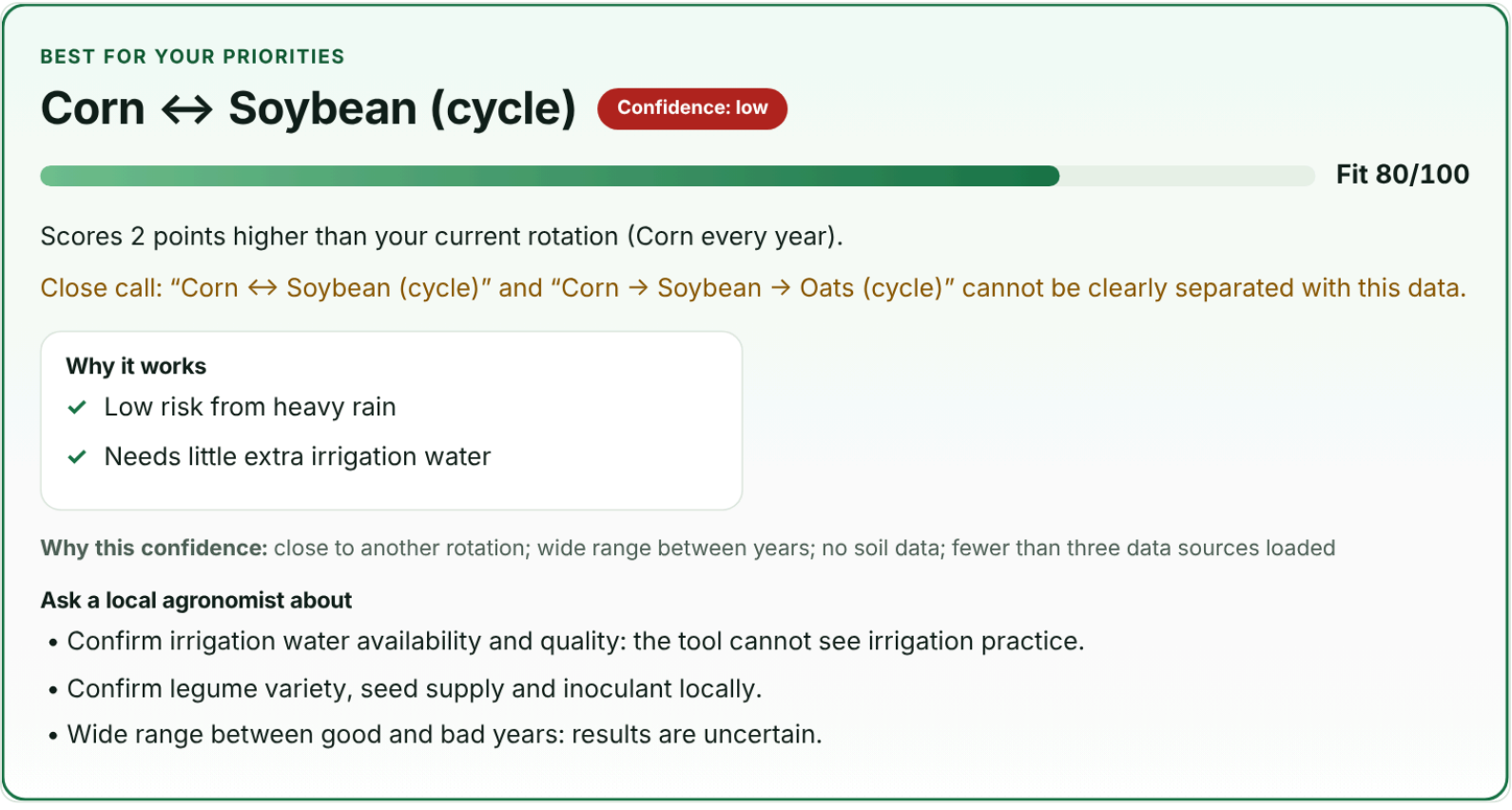
1. **Where is your farm?** Search for a town (suggestions appear as you type), press "Use my location", tap the map, or choose a quick pick. A draggable pin shows the chosen point and a bar under the map shows its status.
2. **What matters most to you?** Three sliders (water saving, income and yield risk, soil building), each with a one-line explanation, plus three big buttons for how you water your crops: not sure, irrigated or rain-fed only.
3. **Our suggestion.** The best option, a "Fit /100" meter, "Why it works" and "Watch out for" in everyday words, a confidence level with its reasons, and what to ask a local agronomist. Every option is listed below; the technical evidence is folded under "Evidence and details".
4. **More details and tools.** Field profile, calendar, what-if tests, your own crop rules, your own rotation, and place comparison.



Landing section (light theme).



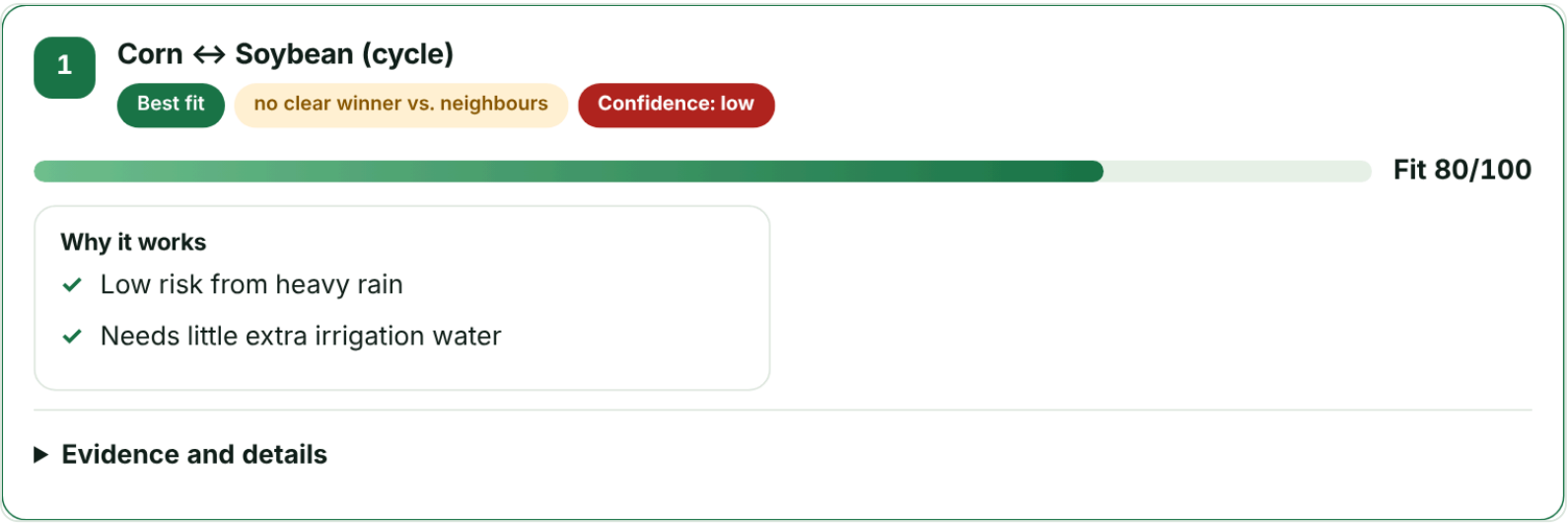
Step 2: priorities and water source.



Step 3: the suggestion for Ames, Iowa, using cached real NASA POWER weather. This file has no soil data, so the app itself reports low confidence and says why. It does not hide the gap.

2. What a user sees (continued)

Every option gets its own card with the same plain layout. Technical detail (the range across years, the six scores, benefits and risks, data gaps and the agronomist checklist) is folded under "Evidence and details" so it is there for anyone who wants it, but never in the way.



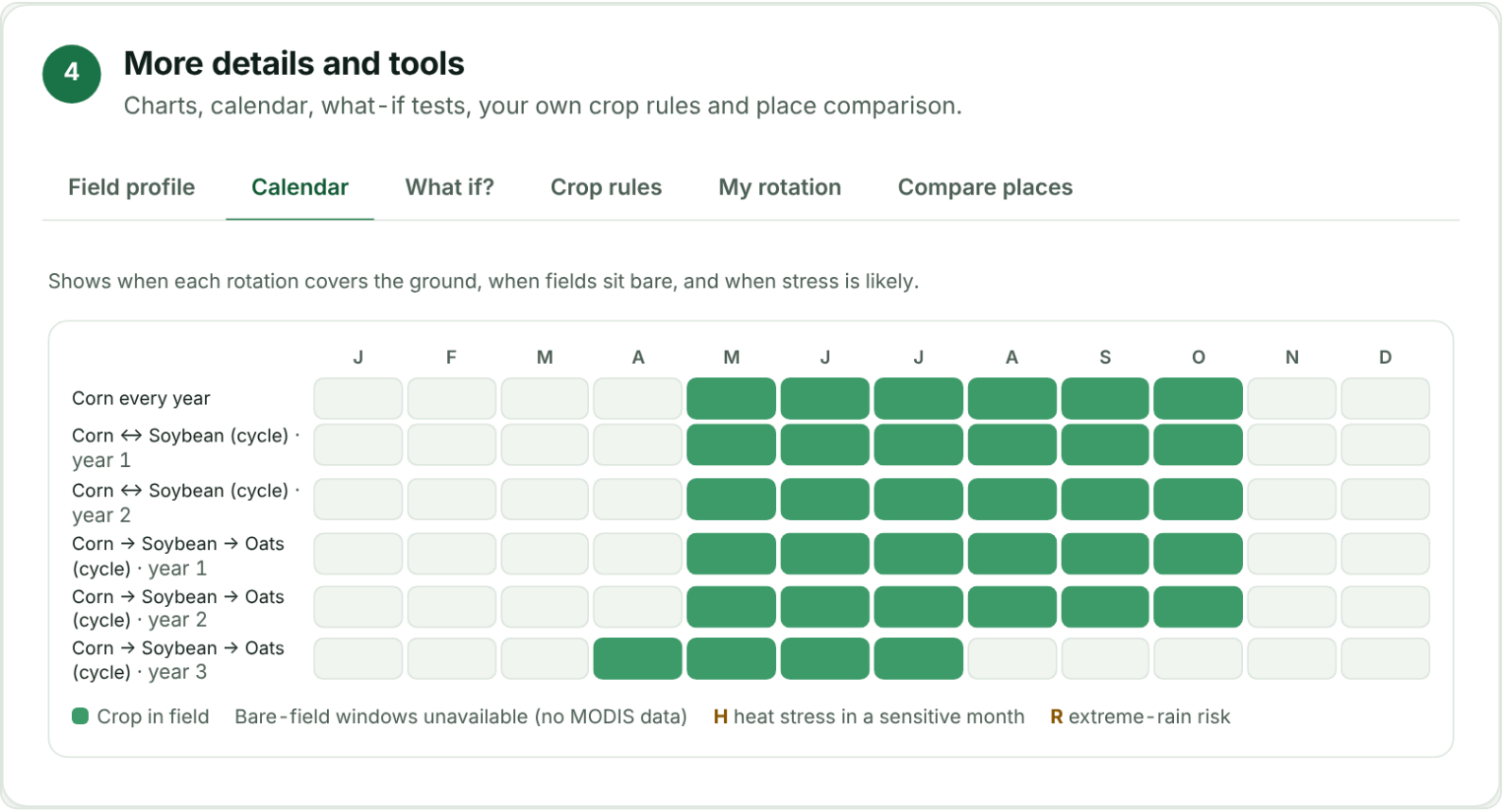
One option card: fit meter, strengths and watch-outs, with the evidence folded away.

The language switcher at the top right changes every word on the page, including the explanations, risks, data gaps and crop names. Urdu and Arabic also mirror the whole layout, so the page reads naturally right to left.



The same suggestion in Urdu (right-to-left layout).

Step 4 holds the optional tools. The calendar below shows, for each rotation, when the field is covered by a crop, when it sits bare, and when heat stress or heavy rain is likely.



Step 4: the tools panel (calendar tab).

3. Everything we added

Choosing a place

FEATURE	WHAT IT DOES
Map click and draggable pin	Click anywhere or drag the pin; data loads automatically. The map no longer jumps under your cursor, and quick-pick markers hide when you zoom in so you can drop a pin anywhere.
Search with suggestions	Place suggestions as you type, returned in your chosen language (Open-Meteo geocoding, free, no key).
Use my location	Asks the browser for permission; handles denied, unavailable, timed-out and insecure-page cases with clear messages.
9 quick picks	Lahore, Islamabad, Rawalpindi, Ludhiana, Dhaka, Nairobi, Londrina, Toowoomba and Ames. Places with a cache file open instantly; others load live.
Anywhere on Earth	Live NASA POWER and SoilGrids for any point. If the pin is on a city (SoilGrids has no data there) the nearest farmland is used and the distance is shown.

Analysis

FEATURE	WHAT IT DOES
Summary and confidence	Best option, comparison with the current rotation, confidence with reasons, and a per-card confidence badge.
Water source	Irrigated farms are scored on how much water is needed; for rain-fed farms a water gap counts as yield risk and the water-saving slider switches off. Each option shows its typical water gap in mm a year.
What-if tests	Rainfall change, and warming of +0 to +4 °C. Warming re-counts heat and frost days from real daily data, not invented factors. The app also reports the point where the top choice flips.
Crop rules editor	A local expert can change water need, sowing and harvest months and pH range per crop, and record a source. The ranking updates live.
My rotation	Build your own rotation (up to 3 years, 2 crops each); it joins the ranked list and the calendar.
Compare places	Two places side by side under the same priorities, water source and rules.
Field profile and calendar	Rainfall chart with ranges, heat and frost days, soil table with uncertainty, and a calendar showing when fields are covered, bare or stressed.
More rotations	12 in total; Punjab places now compare 7, including rice + wheat, cotton + wheat and sugarcane.

Trust and honesty

FEATURE	WHAT IT DOES
Data-used chips	Show which of POWER, SoilGrids, MODIS, SMAP and IMERG actually fed the result.
Data gaps and agronomist flags	Every card lists what the data could not see and what to confirm locally (irrigation, soil salinity, seed and inoculant, and so on).
Source badges	Each crop rule is marked Sourced, Partly sourced or Team estimate, with links to the documents.

3. Everything we added (continued)

Sharing, access and quality

FEATURE	WHAT IT DOES
Copy link	The whole view (place, priorities, water source, what-ifs, edited rules, own rotation, tab, language) lives in the address. Opening the link in a fresh browser restores it exactly. Hostile links are sanitised.
Print / save as PDF	A clean report with every tab, the link and a disclaimer.
Installable and offline	A service worker caches the app and cached places so they work with the server off.
10 languages	English, Urdu, Hindi, Arabic, Bengali, Spanish, French, Portuguese, Swahili and Chinese; 354 text strings each. Urdu and Arabic mirror the whole layout. **Written by AI; not yet reviewed by native speakers.**
Accessibility and phones	Automated axe-core audit: no violations in light, right-to-left and phone layouts across all tabs. Large tap targets and no sideways scrolling on a 390 px phone.

Data pipeline (Python)

PIECE	WHAT IT DOES
`run_all.py` / `build-data.bat`	**One command** that installs what it needs, asks for your NASA login (password hidden), fetches everything, waits for NASA, builds the files, and offers to push to GitHub. Safe to re-run.
`fetch_power.py`	20 years of daily NASA POWER data for each place.
`fetch_soilgrids.py`	SoilGrids soil with uncertainty; finds the nearest farmland when the pin is on a city; three places at a time because the server is slow.
`appeears.py`	Submits, waits for, and downloads MODIS NDVI and SMAP jobs; resubmits automatically when the list of places changes.
`build_cache.py`	Builds one small file per place, splits multi-place NASA files by place, stores warming counts, and lists every missing source as a gap.

Crop calendars and water needs: where the numbers come from

RULE	SOURCE (OPENED AND READ BY US UNLESS NOTED)	STATUS
Seasonal water need, 9 crops	FAO Irrigation Water Management manual (Brouwer & Heibloem), Table 5; midpoint of each range	Sourced
Punjab wheat calendar	FAO/GIEWS Foodcrops & Shortages 02/02 and FEWS NET Pakistan report, May 2026	Sourced
Iowa corn, soybean, oats	USDA NASS Agricultural Handbook 628, Iowa tables	Sourced
Punjab rice, cotton, sugarcane	FEWS NET kharif window (Apr–Jul planting, Sep–Nov harvest); exact months are ours	Partly
Punjab canola / mustard	Punjab Agriculture Department variety table (sown 1–15 Oct; 130-day variety)	Partly
Punjab chickpea, mungbean	Pakistani agronomy notes seen only as search summaries; not opened	Partly
Heat, frost, rain sensitivity, pH, carbon effect; other regions	Team rules of thumb	Estimate

4. What we did, in order

STAGE	WHAT HAPPENED
1. Plan	Read your proposal; chose a rule-based, evidence-first design; created the repository structure.
2. Foundations	Built the scoring engine, the data scripts, the first web app, tests, and clearly labelled placeholder data so the app could run before real data existed.
3. Real data, first attempts	You ran POWER, SoilGrids and AppEEARS on your machine; we fixed the SMAP product version (.008) and the Earthdata variable names.
4. Any location	Added live NASA POWER and SoilGrids lookups in the browser so any point works.
5. Two redesigns	Dark workspace with tabs, then a light four-step page in plain language after your feedback.
6. Ten languages	Translation engine, right-to-left layout, and 354 strings per language with automated completeness checks.
7. Map and search	Replaced map tiles that now need an API key, added suggestions, GPS, a status bar, a draggable pin, and fixed clicks that seemed to do nothing.
8. Big feature batch	Summary and confidence, water source, temperature what-if, crop rules editor, compare places, share link, print report, offline use, accessibility fixes.
9. Data discoveries	Found why soil was empty (SoilGrids has no data under cities), fixed a bug that would have blended all places' satellite data, found that your NASA login cannot be used from the cloud session, and wrote the one-command builder.
10. Sources and rotations	Read FAO, FEWS NET, USDA and Punjab agriculture documents; added sourced calendars and water needs, three Punjab rotations, and the "My rotation" builder.

Problems we found and fixed along the way

- Soil came back empty: SoilGrids has no values under built-up land. The pipeline now uses the nearest farmland and records the distance, and the app shows it.
- A multi-place NASA file would have been averaged into one blend; it is now split by place and covered by a test.
- Map tiles showed "API key required"; switched to OpenStreetMap tiles.
- My old wheat water figure was the bottom of the FAO range, which flattered wheat; every sourced crop now uses the midpoint.
- Screenshots and tests caught layout bugs: sideways scrolling on phones, a hidden button showing too early, a style clash in the crop-rules table, a heading-order issue, and a stale sources table.

Quality checks

CHECK	RESULT
Unit tests (scoring, live data, translations, places, crop sources, cache builder, NASA job bookkeeping)	**All 7 suites pass**
Translations: every language has every text and every placeholder	**10 languages × 354 strings**
Browser sweep: every language, desktop and phone, every tool tab	**120 of 120**
Accessibility audit (axe-core), light, right-to-left, phone, all tabs	**No violations**
End-to-end flows: search, map, drag, retry, sliders, what-if, rules, compare, share link, print, GPS, offline	**Pass**

Caveat: the browser tests use simulated NASA, SoilGrids and geocoding responses because the build environment cannot reach those services from a page. Your own browser test confirmed that the place suggestions and NASA POWER work live; SoilGrids is slow or unreliable for city points.

5. Honest limits

- **It compares options; it does not forecast yield.** Results describe the surrounding area (NASA POWER is about 50 km), not one field.
- **Real data is real but approximate.** Rainfall can differ from a local weather station by 20–30%. SoilGrids is a 250 m model: for Faisalabad it shows about 1% organic matter, which is higher than the "below 0.5%" in newspaper reports. Say "reports say" for that figure.
- **Crop rules are partly team estimates.** Only the items marked Sourced come from documents we read. Chickpea and mungbean windows come from secondary sources we did not open.
- **Places typed or clicked get fewer data sources** (no MODIS or SMAP), so their confidence is capped at medium.
- **Translations are AI-written** and need a fluent reviewer, especially Urdu.
- **Out of scope by design:** training an AI model (no labelled outcome data, and it would be harder to defend), prices and profit (no data), and field-level yield prediction.

6. What is left to do

- ☐ **Build the real data:** run `python pipeline/run_all.py` (or double-click `build-data.bat`), enter your Earthdata login when asked, and let it push the result.
- ☐ **Deploy a public link:** turn on GitHub Pages for the `app` folder and test it in a private window.
- ☐ **Test one search in your browser** and tell us if anything fails (press F12, Console tab).
- ☐ **Check the sources:** open the links in the app's "Crop rule sources" table and confirm the months.
- ☐ **Native-speaker review** of at least the Urdu text.
- ☐ **Read the 2026 challenge pages** (we could not open them) and check the submission limits.
- ☐ **Record the 30-second video** or build 7 slides, using the script and checklist in `docs/SUBMISSION.md`. Name only the NASA datasets actually loaded for the place you demo.
- ☐ **Submit** well before 11:59 p.m. local time on Sunday 15 November.
- ☐ Your unfinished list item “3.” from earlier: tell us what it was.

7. How to run it

```
git pull
python -m http.server 8000 --directory app      # then open http://localhost:8000
python pipeline/run_all.py                       # builds the real data (Windows: python, not python3)
```

Tests: `node tests/scoring.test.js`, `node tests/i18n.test.js`, `node tests/crops.test.js`, `node tests/places.test.js`, `node tests/live.test.js`, `python tests/test_build_cache.py`, `python tests/test_appeears.py`. Full details are in `README.md`; the submission kit is `docs/SUBMISSION.md`.

Data credits: NASA POWER, NASA MODIS, NASA SMAP, NASA GPM IMERG; SoilGrids (ISRIC, not NASA); crop information from FAO, FEWS NET, USDA NASS and the Punjab Agriculture Department. AI tools (Claude) helped write the code and this report; the team reviews and owns the choices.